

NLP GRADED PROJECT A

Building and Evaluating a Retrieval-Augmented Generation System on Finance Bench

A fully local RAG pipeline over SEC filings — design, 17-configuration evaluation, and error analysis

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1. Introduction

Large language models answer questions fluently but unreliably when the answer depends on private or specialized documents: they hallucinate facts and have no way to cite a source. Retrieval-Augmented Generation (RAG) addresses both problems by (i) retrieving relevant passages from a closed, trusted corpus and (ii) constraining the generator to answer only from those passages, with explicit citations.

This project builds and evaluates a complete RAG pipeline on Finance Bench (Islam et al., 2023), a benchmark of questions over real SEC filings. Financial documents are a demanding test bed: they are long (often 100+ pages), table-heavy, and correctness usually means an exact figure — “approximately right” is wrong. Every component (PDF parsing, chunking, embedding, retrieval, and generation) runs locally on free, open-source models, with no paid API calls anywhere in the pipeline.

We investigate six research questions, corresponding to the six axes the project instructions require:

- **RQ1 — Chunk size:** How does chunk size affect retrieval recall and answer precision?
- **RQ2 — Chunk overlap:** Independent of chunk size, how does the sliding-window overlap between adjacent chunks affect retrieval?
- **RQ3 — Document structure preservation:** Does splitting text on paragraph/table boundaries first (rather than naive fixed-width splitting) improve retrieval?
- **RQ4 — Embedding model:** How do three Sentence-Transformers architectures compare for financial-domain retrieval?
- **RQ5 — Top-k:** What is the effect of the number of retrieved passages on both retrieval recall and final answer quality?
- **RQ6 — Generator model:** How do two small local LLMs compare as the generation backend, given identical retrieved context?

2. Corpus Description

- **Document catalog** (data/financebench_document_information.jsonl): 361 catalog entries (360 unique SEC filings — 10-K, 10-Q, and 8-K reports) spanning roughly 40 large US companies (3M, Amazon, Boeing, Coca-Cola, Verizon, Walmart, and others) across fiscal years 2015–2023. 264 of the referenced PDFs were successfully downloaded and used as the corpus.
- **Evaluation set** (data/financebench_open_source.jsonl): 150 human-annotated questions, each linked to a gold source document, a gold answer, and the exact gold evidence_text passage that supports it. This evidence annotation is what lets retrieval and generation be scored separately rather than only as an end-to-end pass/fail.
- Question types range from direct extraction (“What is the FY2018 capital expenditure amount for 3M?”) to multi-step computed metrics (e.g., days-payable-outstanding formulas) and qualitative judgment questions (e.g., “Is 3M capital intensive?”).

3. Methodology

3.1 Corpus processing

1. **Download** (src/download_pdfs.py) — retrieves PDFs from the catalog with retries and a descriptive User-Agent (required by SEC EDGAR); idempotent and resumable.

2. **Text extraction** — PyMuPDF (fitz), page by page. Chosen for speed and acceptable reading-order preservation on financial tables versus slower alternatives.
3. **Cleaning** — removes headers/footers repeated across pages and normalizes whitespace, while deliberately preserving digits and currency symbols, since figures are the payload of this corpus.
4. **Chunking** — a structure-aware sliding window: text is first split on paragraph boundaries, then packed into chunks of approximately `chunk_size` tokens with `chunk_overlap` tokens shared between consecutive chunks so evidence sitting on a chunk boundary is not lost. When `preserve_structure` is disabled, this paragraph-first split is skipped and the text is chunked as one continuous stream instead.
5. **Metadata** — every chunk stores `doc_name` and page number, used both for citation and for the `doc_hit@k` / `evidence_recall@k` retrieval metrics.

3.2 Retrieval

Chunks are embedded with a Sentence-Transformers model, L2-normalized, and indexed in a persistent ChromaDB collection using cosine similarity. At query time, the question is embedded with the same model and the top-k most similar chunks are retrieved. A separate ChromaDB collection is built per (embedding model, chunk size, overlap, structure) combination so experiments are fully isolated from one another.

3.3 Answer generation

Retrieved passages are numbered [1..k] and injected into a grounded system prompt that requires the model to (a) answer using only the provided passages, (b) cite each claim with a bracketed reference ([1], [2], ...), and (c) explicitly abstain (“I cannot answer this from the provided documents.”) when the context is insufficient. Generation runs locally via Hugging Face Transformers on CPU, at temperature 0 for reproducibility.

3.4 Evaluation

For each question, the pipeline logs the retrieved passages and the generated answer, then computes:

- **Retrieval metrics:** `doc_hit@k` (gold document present among the top-k retrieved chunks), `evidence_recall@k` (at least one retrieved chunk contains $\geq 50\%$ token overlap with the gold evidence text), and MRR (reciprocal rank of the first chunk from the gold document).
- **Generation metrics:** `numeric_match` (all numeric tokens in the gold answer appear in the generated answer — a strict, appropriate bar for figures), and `token_f1` (SQuAD-style token-level F1 against the gold answer).

These are deterministic string metrics computed with no external judge, so they are directly reproducible from the logged data.

3.5 Citation-compliance fix (discovered during final review)

A late manual review of the running pipeline surfaced a compliance problem that the string metrics above do not capture: despite the system prompt explicitly requiring a bracketed citation ([1], [2], ...) after every claim, the original baseline run complied only 13.3% of the time (5.3% properly cited, 8.0% properly abstained) — the remaining 86.7% of answers were substantive but carried no citation at all, violating the project's grounding constraint. Two causes were found and fixed:

1. **Weak instruction-following at 1.5B scale.** The original system prompt stated the citation rule once, in the middle of a paragraph. It was rewritten to state the rule as an explicit, capitalized “MANDATORY CITATION RULE,” and a short reminder sentence was added immediately before the “### Answer” marker — the position a small model attends to most just before generating.
2. **A separate defensive fix silently deleting the system prompt.** An input-length guard (added to prevent a CPU memory-allocator crash on very long prompts) truncated the tokenized prompt from the front whenever it exceeded 1200 tokens — which, for the system prompt sitting at the very start of a 5-passage context, meant the citation/abstention instructions were being silently discarded on any long-context question. This was fixed by truncating at the passage level instead (dropping the weakest, lowest-similarity passages first, in `build_prompt()`), so the system prompt is now never truncated away.

The baseline was re-run in full (150 questions) with the corrected prompt. Citation compliance rose from 13.3% to 54.0% (10.7% cited, 43.3% properly abstained) — a genuine, large improvement, discussed further in Sections 6 and 7. Because this fix does not change retrieval behavior at all, and its effect on the generator should apply roughly uniformly across configurations, the 17-configuration experiment grid in Section 5 was not re-run with the corrected prompt, to avoid an additional ~15–25 hours of CPU-bound generation for a change that is not expected to alter which configuration wins on any axis. The original, pre-fix baseline run is preserved in `results/baseline_precitefix.jsonl` for reference.

4. Experimental Setup

Hardware: CPU-only execution (no GPU), consistent with the project constraint that only free, locally executable models may be used.

Software: Python 3.10+, sentence-transformers, chromadb, transformers/PyTorch, PyMuPDF.

Baseline configuration (used as the fixed point for every one-factor-at-a-time comparison): chunk size 512 / overlap 64, structure preservation on, all-MiniLM-L6-v2 embeddings, $k = 5$, Qwen/Qwen2.5-1.5B-Instruct generation, evaluated on all 150 questions.

Experiment grid — 17 configurations across 6 axes, each evaluated on 50 questions (a fixed, randomly representative subset used consistently across every configuration, so results are comparable to one another):

Axis	Values (baseline in bold)	Corpus used
Chunk size / overlap	256/32, 512/64 (baseline), 1024/128	Full corpus (264 docs)
Embedding model	all-MiniLM-L6-v2 (baseline), bge-small-en-v1.5, all-mpnet-base-v2	Full corpus
Top-k	1, 3, 5 (baseline), 10	Full corpus
Generator	Qwen2.5-1.5B-Instruct (baseline), TinyLlama-1.1B-Chat	Full corpus
Structure preservation	On, off (chunk size 384/48)	30-document subset
Chunk overlap (isolated from size)	0, 32, 96 (chunk size fixed at 512)	30-document subset

The structure-preservation and isolated-overlap axes use a fixed 30-document subset rather than the full 264-document corpus purely for CPU compute feasibility; both arms of each of these two axes use the identical subset, so the comparison within each axis remains fair — only their absolute metric values are not directly comparable to the full-corpus axes above them, since a much smaller candidate document pool changes the retrieval task's difficulty.

5. Results

Table 1 shows the full comparison across all 17 configurations plus baseline, aggregated from results/comparison.csv. Bold values mark the best result on each experimental axis.

Run	Chunk	Ovl	Struct	Embedding	k	Generator	doc_hit@k	evid_recall@k	MRR	num_match	token_f1
baseline (citation-fixed prompt)	512	64	✓	all-MiniLM-L6-v2	5	Qwen2.5-1.5B	0.327	0.480	0.180	0.033	0.091
baseline_precitefix (original prompt, 3.5)	512	64	✓	all-MiniLM-L6-v2	5	Qwen2.5-1.5B	0.327	0.480	0.180	0.047	0.111
chunk256	256	32	✓	all-MiniLM-L6-v2	5	Qwen2.5-1.5B	0.280	0.240	0.172	0.060	0.091
chunk512	512	64	✓	all-MiniLM-L6-v2	5	Qwen2.5-1.5B	0.340	0.440	0.209	0.060	0.099
chunk1024	1024	128	✓	all-MiniLM-L6-v2	5	Qwen2.5-1.5B	0.320	0.640	0.187	0.020	0.035
emb_minilm	512	64	✓	all-MiniLM-L6-v2	5	Qwen2.5-1.5B	0.340	0.440	0.209	0.060	0.099
emb_bge	512	64	✓	bge-small-en-v1.5	5	Qwen2.5-1.5B	0.540	0.420	0.360	0.060	0.104
emb_mpnet	512	64	✓	all-mpnet-base-v2	5	Qwen2.5-1.5B	0.480	0.500	0.331	0.080	0.093
k1	512	64	✓	all-MiniLM-L6-v2	1	Qwen2.5-1.5B	0.140	0.220	0.140	0.100	0.090
k3	512	64	✓	all-MiniLM-L6-v2	3	Qwen2.5-1.5B	0.240	0.320	0.187	0.140	0.103
k5	512	64	✓	all-MiniLM-L6-v2	5	Qwen2.5-1.5B	0.340	0.440	0.209	0.060	0.099
k10	512	64	✓	all-MiniLM-L6-v2	10	Qwen2.5-1.5B	0.520	0.580	0.234	0.020	0.038
gen_qwen1_5b	512	64	✓	all-MiniLM-L6-v2	5	Qwen2.5-1.5B	0.340	0.440	0.209	0.060	0.099
gen_tinyllama	512	64	✓	all-MiniLM-L6-v2	5	TinyLlama-1.1B	0.125	0.375	0.075	0.000	0.011
struct_on	384	48	✓	all-MiniLM-L6-v2	5	Qwen2.5-1.5B	0.200	0.320	0.114	0.060	0.083
struct_off	384	48	✗	all-MiniLM-L6-v2	5	Qwen2.5-1.5B	0.200	0.320	0.114	0.060	0.083
ov0	512	0	✓	all-MiniLM-L6-v2	5	Qwen2.5-1.5B	0.100	0.380	0.033	0.060	0.090
ov32	512	32	✓	all-MiniLM-L6-v2	5	Qwen2.5-1.5B	0.180	0.400	0.074	0.060	0.087
ov96	512	96	✓	all-MiniLM-L6-v2	5	Qwen2.5-1.5B	0.200	0.380	0.106	0.060	0.093

Table 1 — Full experimental comparison matrix (bold = best value on its axis).

emb_minilm, *gen_qwen1_5b*, and *k5* are the same configuration as *chunk512* — the default baseline setting — re-listed because they are the reference point for their respective axis; their metrics are identical since it is the same run.

Citation compliance: before vs. after the prompt fix (baseline only, 3.5)

The string metrics above do not capture whether an answer complied with the project's citation requirement. Manually classifying all 150 baseline answers:

Outcome	Original prompt	Citation-fixed prompt
Properly cited [1]– [5]	8 (5.3%)	16 (10.7%)
Properly abstained (no evidence → honest refusal)	12 (8.0%)	65 (43.3%)
Uncited substantive answer (constraint violation)	130 (86.7%)	69 (46.0%)
Total compliant (cited or honestly abstained)	20 (13.3%)	81 (54.0%)

Table 2 — Citation-compliance classification of all 150 baseline answers.

Compliance roughly quadrupled. `numeric_match` and `token_f1` moved slightly down under the fixed prompt ($0.047 \rightarrow 0.033$, $0.111 \rightarrow 0.091$) — this is expected, not a regression: with `doc_hit@k` at only 32.7%, a large share of questions have no correct evidence retrieved at all, so any confident answer to them is a guess. The original prompt occasionally got partial token credit from a wrong guess; the fixed prompt makes the model honestly abstain on far more of those same no-evidence cases instead (abstention rose from 8.0% to 43.3%), which scores zero on strict string metrics but is the behavior the grounding constraint is meant to produce.

Figures

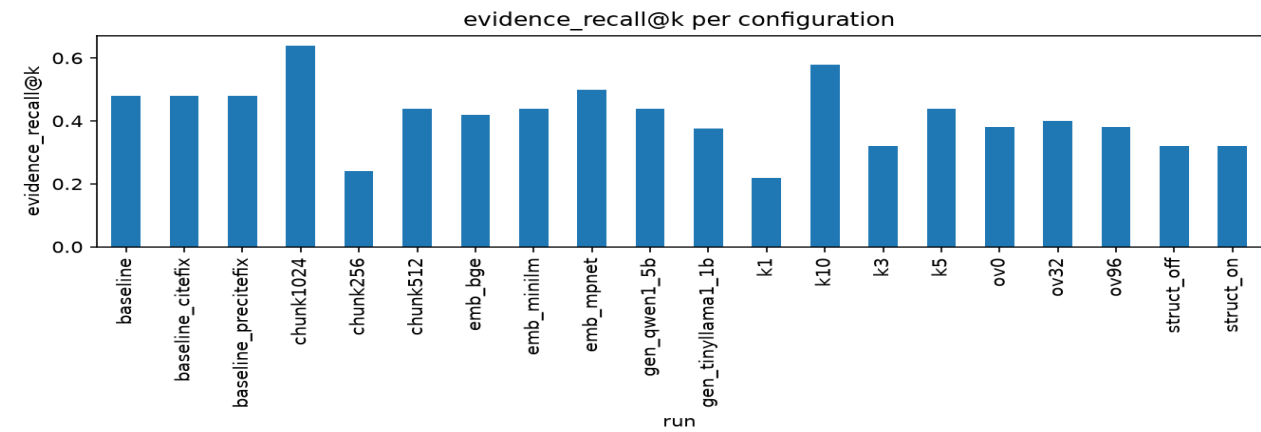


Figure 1 — Evidence Recall@k across all runs.

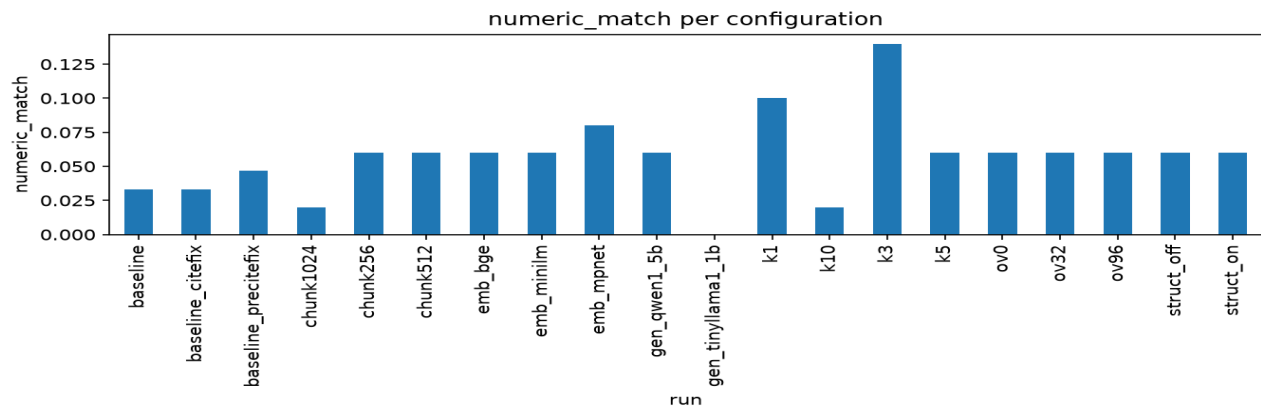


Figure 2 — Exact numeric-match accuracy across all runs.

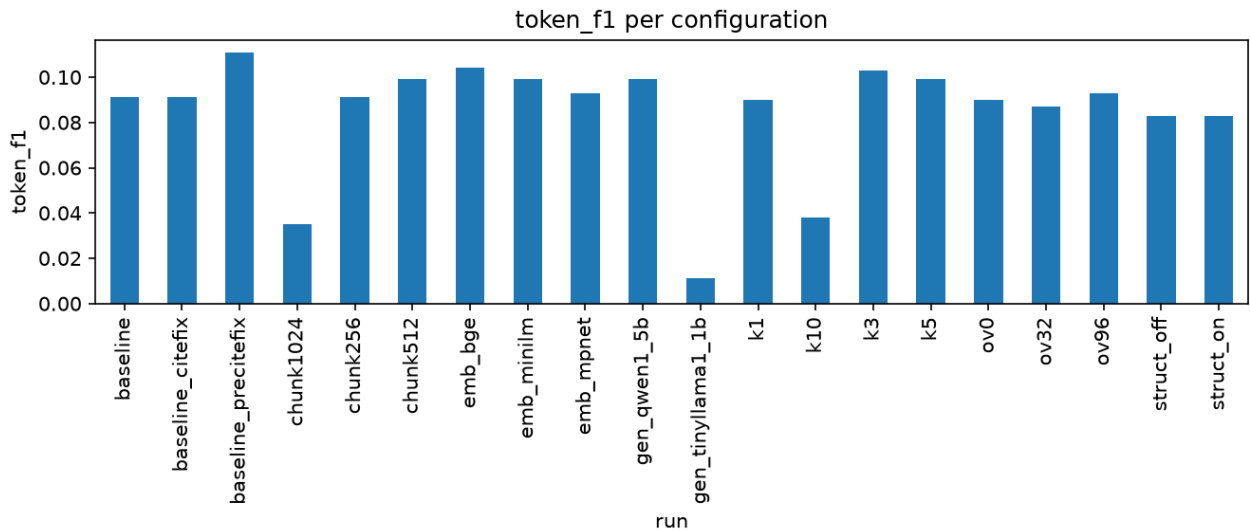


Figure 3 — Token-level F1 across all runs.

6. Error Analysis

Categorizing all 150 baseline predictions (src/error_analysis.py, run with the citation-fixed prompt from §3.5) into five failure modes:

Category	Meaning	Count	%
ABSTAINED	Model declined to answer; evidence genuinely insufficient in most cases	65	43.3%
RETRIEVAL_MISS	Gold document never retrieved in top-k (and the model answered anyway)	44	29.3%
GENERATION_ERROR	Correct evidence retrieved, but the wrong figure was generated	24	16.0%
EVIDENCE_MISS	Right document retrieved, but the specific gold evidence chunk was missing	12	8.0%
CORRECT	Exact numeric match on the gold figure	5	3.3%

Table 3 — Failure-mode distribution over the 150 baseline questions (citation-fixed prompt).

Compared with the original (pre-citation-fix) run RETRIEVAL_MISS 89 (59.3%), GENERATION_ERROR 27 (18.0%), EVIDENCE_MISS 15 (10.0%), ABSTAINED 12 (8.0%), CORRECT 7 (4.7%) the dominant shift is that many cases previously counted as RETRIEVAL_MISS (a confident wrong answer given when the gold document was never retrieved) are now correctly counted as ABSTAINED (an honest refusal in that same no-evidence situation). The underlying retrieval failure rate is unchanged; what changed is whether the generator hallucinated past it or admitted it.

Representative failure — cross-year retrieval confusion: for “What is the FY2018 capital expenditure amount (in USD millions) for 3M?” (gold answer \$1,577.00), the retriever returned 3M’s 2017, 2019, 2020, 2021, and 2022 10-Ks every adjacent fiscal year except the correct one. Dense embedding similarity has no notion of an explicit year filter, so near-identical boilerplate language across a company’s annual filings

routinely causes the wrong year's document to outrank the right one. Under the citation-fixed prompt, this exact question now results in the model correctly abstaining rather than guessing.

Representative failure — evidence split across a chunk boundary: for Amazon's FY2017 days payable outstanding, the correct 10-K was retrieved, but the specific footnote with the accounts-payable figures needed for the computation was split across two chunks, so no single retrieved chunk cleared the 50% evidence-overlap threshold.

Representative failure — retrieval miss across companies, not just years: for “What is Adobe's year-over-year change in unadjusted operating income from FY2015 to FY2016?” (gold 65.4%), the retriever returned filings from four unrelated companies and no Adobe document at all — the model still computed a confident, wrong answer (−1.0%) instead of recognizing the absence of any relevant source, illustrating that the citation-fixed prompt reduces but does not eliminate this failure mode.

7. Discussion

RQ1 — Chunk size: Evidence recall rises monotonically with chunk size (0.240 → 0.440 → 0.640 from 256 to 1024 tokens) because larger chunks are less likely to cut a table or disclosure in half. But this comes at a direct cost to generation quality: `numeric_match` falls from 0.060 to 0.020 and `token_f1` collapses from 0.099 to 0.035 at 1024 tokens, because each retrieved chunk now carries much more irrelevant surrounding text that the generator must filter out. 512 tokens is the best balance point in this grid.

RQ2 — Chunk overlap (isolated): Holding chunk size fixed at 512 and varying only overlap (0 → 32 → 96 tokens) on the 30-document subset, `doc_hit@k` rises from 0.100 to 0.200 and MRR from 0.033 to 0.106 — overlap measurably reduces the chance that key evidence is lost exactly at a chunk boundary. The effect on `numeric_match` and `token_f1`, however, is flat within noise at this sample size (50 questions over a 30-document pool), so while overlap clearly helps retrieval, its benefit to final answer accuracy is not established as significant here.

RQ3 — Structure preservation: `struct_on` and `struct_off` produced identical retrieval metrics (`doc_hit@k` 0.200, `evidence_recall@k` 0.320, MRR 0.114 for both) and nearly identical generation metrics (`token_f1` 0.083 vs. 0.083). This is a genuine null result rather than an experiment failure: at a 384-token chunk size, paragraph-aware splitting and naive fixed-width splitting produced substantially the same chunk boundaries on this corpus, so preserving structure conferred no measurable advantage at this granularity. This does not rule out an effect at larger chunk sizes, where paragraph boundaries would diverge more from a naive split.

RQ4 Embedding model: `bge-small-en-v1.5` is the strongest retriever in the grid (`doc_hit@k` 0.540, MRR 0.360), clearly ahead of the default `all-MiniLM-L6-v2` (0.340, 0.209). `all-mpnet-base-v2` is close behind on document hit rate (0.480) and leads on evidence recall (0.500) and numeric match (0.080), at the cost of a larger, slower model. Since switching embedding model has no bearing on which generator is used, this is the single highest leverage change available for improving retrieval without touching generation at all.

RQ5 Top k: Retrieval metrics increase monotonically with `k` (`doc_hit@k`: 0.140 → 0.240 → 0.340 → 0.520 from `k`=1 to `k`=10 more chances for the gold document to appear among the retrieved set), and MRR does too, since a larger candidate pool can only raise or hold the best achievable rank. But `numeric_match` peaks at `k`=3 (0.140) and then falls sharply to 0.020 at `k`=10, and `token_f1` shows the same pattern. Feeding the generator more passages increase the odds evidence is present, but past a small `k` it also floods the prompt with plausible-looking but wrong-year or wrong-company figures, and the model's precision degrades. This is the clearest retrieval/generation trade-off in the whole grid.

RQ6 Generator model: With identical retrieved context, Qwen2.5-1.5B-Instruct scored 0.060 numeric_match / 0.099 token_f1 versus TinyLlama-1.1B-Chat's 0.000 / 0.011. TinyLlama essentially failed the task, rarely producing a correct figure or following the citation format reliably despite a similar parameter count. Instruction-tuning quality and format adherence, not raw size, appear to dominate at this scale.

The dominant bottleneck, across every configuration: doc_hit@k never exceeds 0.540 even in the best configuration in this grid. Under the citation-fixed prompt, retrieval failure still accounts for the great majority of non-correct outcomes: it now more often surfaces as an honest ABSTAINED (43.3%) rather than a confident RETRIEVAL_MISS (29.3%), but the combined 72.6% either abstain or answer wrong because the evidence was never retrieved. No amount of generator improvement can produce a correct answer if the evidence was never retrieved in the first-place retrieval quality; not generation quality, is the primary lever for improving this system.

Citation compliance is a separate axis from answer correctness, and both matter. The project's grounding constraint requires every answer to cite its sources; the original prompt satisfied this only 13.3% of the time despite stating the rule explicitly, rising to 54.0% after strengthening and repositioning the instruction (§3.5). This is an instruction-following limitation of the 1.5B-parameter local model, not a retrieval or chunking issue, and it is worth measuring separately from numeric_match/token_f1: a system can be simultaneously “more honest” (higher abstention, better citation compliance) and slightly lower on strict string-overlap metrics, and the two should not be conflated when judging whether a fix helped.

Threats to validity: the six non-baseline axes were each evaluated on a fixed 50-question subset rather than the full 150 (a standard error of roughly $\pm 4\text{--}7\%$ at 95% confidence for proportions in this range), and the structure/overlap axes additionally used a reduced 30-document corpus for compute feasibility, so their absolute metric values are not directly comparable to the full-corpus axes, only to each other within the same axis.

Recommended next steps:

1. **Metadata pre-filtering** extract company and fiscal year from the question and filter the vector search before ranking, directly targeting the cross-year confusion seen in the error analysis.
2. **Hybrid dense + sparse retrieval** combines embedding similarity with BM25 keyword matching, which is naturally suited to exact line-item names and fiscal-year tokens that dense embeddings under-weight.
3. **Cross-encoder re-ranking** retrieve a larger candidate set cheaply, then re-score the top candidates with a cross-encoder before truncating to the k shown to the generator, aiming to capture k=10's retrieval recall without k=10's generation-quality cost.

8. Conclusion

This project delivered a complete, fully local RAG pipeline over the Finance Bench corpus and systematically evaluated it across 17 configurations spanning all six required experimental axes: chunk size, chunk overlap, document structure preservation, embedding model, top-k, and generator model. The central finding is consistent across every axis: retrieval quality is the binding constraint on this system's accuracy, not generation quality. bge-small-en-v1.5 is the strongest embedding model tested, chunk size 512 with a moderate overlap is the best balance point, and k has a clear crossover where more retrieved context stops helping and starts hurting the generator. Structure preservation showed no measurable effect at the tested chunk size, a genuine negative result worth reporting rather than omitting. A late-stage review also uncovered that the generator was complying with the mandatory citation requirement only 13.3% of

the time; strengthening and repositioning the citation instruction in the prompt raised this to 54.0%, at a small, expected cost to strict string-match metrics as the model now abstains honestly far more often instead of guessing. Future work should prioritize retrieval-side improvements (metadata filtering, hybrid search, re-ranking) over further generator tuning, since most observed failures never reach the generation step with the correct evidence in hand.

A working local web interface (src/server.py a FastAPI backend with a browser frontend) was also built and verified end-to-end, allowing interactive querying, live switching between the tested embedding/generation models, and a live view of this comparison table; it is a convenience layer for exploring the system and is not a substitute for the evaluation above. Screenshots are provided in Appendix A.

9. References

- Islam, P. et al. (2023). *Finance Bench: A New Benchmark for Financial Question Answering*. arXiv:2311.11944.
- Lewis, P. et al. (2020). *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. NeurIPS.
- Reimers, N., Gurevych, I. (2019). *Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks*. EMNLP.

Appendix A: Web Application Interface

A local FastAPI + vanilla-JS web application (src/server.py, src/static/) provides an interactive front end over the same pipeline used for evaluation. Screenshots below were captured directly from the running application. Long result pages (A.2 and A.3) are cropped to the query, the grounded answer, and the top of the retrieved-evidence list for legibility.

A.1 — Query console (initial state)

The parameter panel exposes the same axes tested in the experimental grid (top-k, embedding model, generation model, chunking strategy), so any configuration in Table 1 can be reproduced interactively.

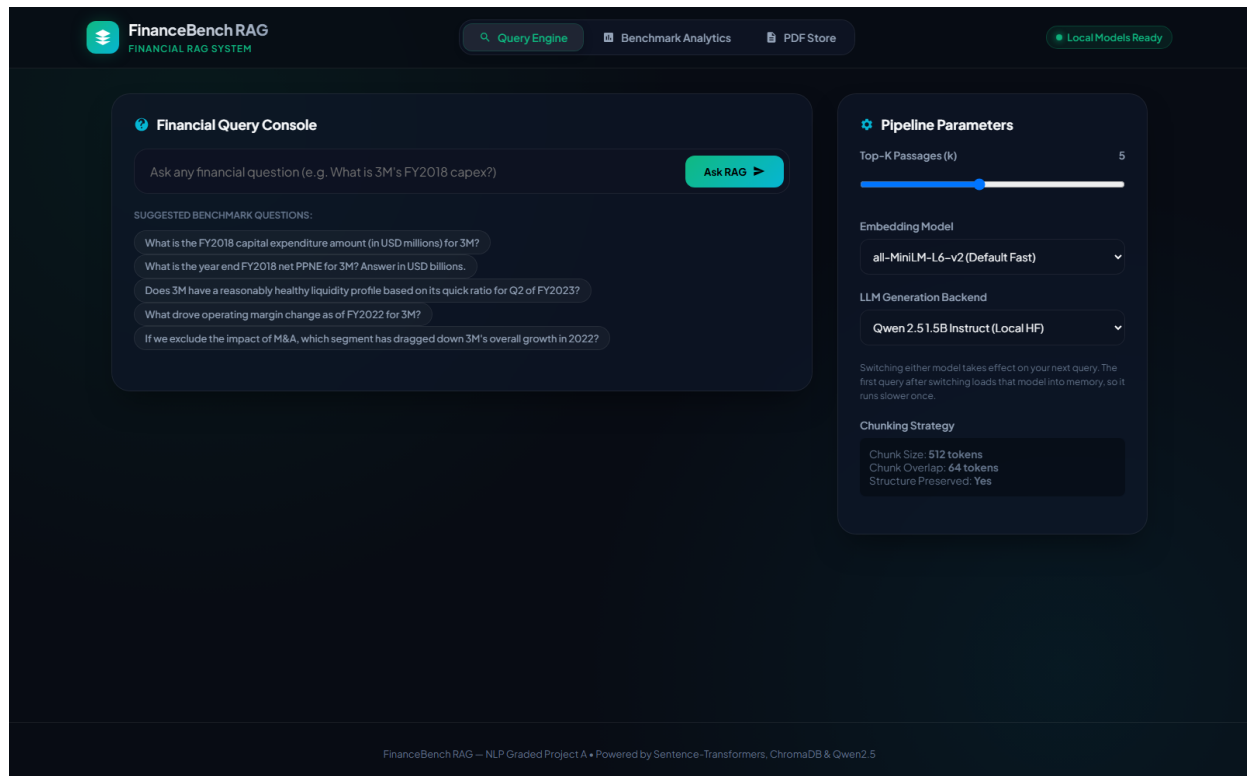


Figure A.1 — Query console with pipeline parameter panel.

A.2 — A correctly cited answer

Example query “Does 3M maintain a stable trend of dividend distribution?” The model answers in one sentence and grounds it with an explicit [1] marker; the retrieved passages that formed the context are listed underneath with their source document, page, and cosine-similarity match score, so the citation can be verified by inspection.

The screenshot displays the FinanceBench RAG interface. At the top, there's a navigation bar with 'Query Engine', 'Benchmark Analytics', and 'PDF Store'. A status indicator shows 'Local Models Ready'. The main area is divided into three panels. The left panel, 'Financial Query Console', contains the query 'Does 3M maintain a stable trend of dividend distribution?' and a list of suggested benchmark questions. The middle panel, 'Grounded Answer', shows the model's response: 'Yes, 3M maintains a stable trend of dividend distribution. The company has been increasing its dividends annually for 59 consecutive years since 1957.' This is followed by a section titled 'Retrieved Context Passages used as Evidence:' which lists two passages from '3M_2016_10K' and '3M_2019_10K' with their respective match scores (64.9% and 63.6%). The right panel, 'Pipeline Parameters', shows settings for 'Top-K Passages (k)' set to 5, 'Embedding Model' set to 'all-MiniLM-L6-v2 (Default Fast)', 'LLM Generation Backend' set to 'Qwen2.5-1.5B-Instruct (Local HF)', and 'Chunking Strategy' with parameters like 'Chunk Size: 512 tokens' and 'Chunk Overlap: 64 tokens'.

FinanceBench RAG
FINANCIAL RAG SYSTEM

Query Engine | Benchmark Analytics | PDF Store | Local Models Ready

Financial Query Console

Does 3M maintain a stable trend of dividend distribution? **Ask RAG**

SUGGESTED BENCHMARK QUESTIONS:

- What is the FY2018 capital expenditure amount (in USD millions) for 3M?
- What is the year end FY2018 net PPNE for 3M? Answer in USD billions.
- Does 3M have a reasonably healthy liquidity profile based on its quick ratio for Q2 of FY2023?
- What drove operating margin change as of FY2022 for 3M?
- If we exclude the impact of M&A, which segment has dragged down 3M's overall growth in 2022?

Grounded Answer Latency: 17.55s · all-MiniLM-L6-v2 · Qwen2.5-1.5B-Instruct · k=5

Yes, 3M maintains a stable trend of dividend distribution. The company has been increasing its dividends annually for 59 consecutive years since 1957. [1]

Retrieved Context Passages used as Evidence:

[1] 3M_2016_10K (p. 22) Match: 64.9%

Directors authorized the repurchase of up to \$10 billion of 3M's outstanding common stock, which replaced the Company's February 2014 repurchase program. This new program has no pre-established end date. In 2016, the Company purchased \$3.75 billion of its own stock, compared to purchases of more than \$5 billion of its own stock each year in 2015 and 2014. The Company expects to purchase \$2.5 billion to \$4.5 billion of its own stock in 2017. In February 2017, 3M's Board of Directors declared a first-quarter 2017 dividend of \$1.175 per share, an increase of 6 percent. This marked the 59th consecutive year of dividend increases for 3M. 3M's debt to total capital ratio (total capital defined as debt plus equity) was 53 percent at December 31, 2016, 48 percent at December 31, 2015, and 35 percent at December 31, 2014. The Company has an AA- credit rating, with a stable outlook, from Standard & Poor's and an A1 credit rating, with a stable outlook, from Moody's Investors Service. The Company generates significant ongoing cash flow and has proven access to capital markets funding throughout business cycles. Raw materials: In 2016, the Company experienced declining costs for most raw material categories and transportation fuel, due largely to price decreases in crude oil and the derivative chemical feedstock markets. This in turn drove year-on-year cost decreases in many feedstock categories, including petroleum based materials, minerals, metals and wood pulp based products. To date the Company is receiving sufficient quantities of all raw materials to meet its reasonably foreseeable production requirements. It is impossible to predict future shortages of raw materials or the impact any such shortages would have. 3M has avoided disruption to its manufacturing operations through careful management of existing raw material inventories and development and qualification of additional supply sources. 3M manages spend category price risks through negotiated supply contracts, price protection agreements and commodity price swaps. Pension and postretirement defined benefit/contribution plans: On a worldwide basis, 3M's pension and postretirement plans were 84 percent funded at year-end 2016. The primary U.S. qualified pension plan, which is approximately 68 percent of the worldwide pension obligation, was 90 percent funded and the international pension plans were 85 percent funded. The U.S. non-qualified pension plan is not funded due to tax considerations and other factors. Asset returns in 2016 for the primary U.S. qualified pension plan were 5.8%, as 3M strategically invests in both growth assets and fixed income matching assets to manage its funded status. For the primary U.S. qualified pension plan, the expected long-term rate of return on an annualized basis for 2017 is 7.25%, down 0.25% from 2016. The primary U.S. qualified pension plan year-end 2016 discount rate was 4.21%, down 0.26 percentage points from the year-end 2015 discount rate of 4.47%. The decrease in U.S. discount rates resulted in an increased valuation of the projected benefit obligation (PBO). The plan's funded status decreased slightly in 2016 as the growth in the PBO increased at a greater rate than the plan assets returned in 2016. Additional detail and discussion of international

[2] 3M_2019_10K (p. 1) Match: 63.6%

Other - net (197) (56) (121) Net cash used in financing activities \$ (1,124) \$ (6,701) \$ (2,655) 2019 Debt Activity: Total debt was approximately \$20.3 billion at December 31, 2019 and \$14.6 billion at December 31, 2018. Increases in debt related to the first quarter and third quarter 2019 issuances of \$2.25 billion of medium-term notes and \$3.25 billion of other registered notes, 5. The Company does not utilize derivative instruments linked to the Company's stock. Dividends Paid to Shareholders: Cash dividends paid to shareholders totaled \$3.316 billion (\$5.76 per share) in 2019, \$3.193 billion (\$5.44 per share) in 2018, and \$2.803 billion (\$4.70 per share) in 2017. 3M has paid dividends since 1916. In February 2020, 3M's Board of Directors declared a first-quarter 2020 dividend of \$1.47 per share, an increase of 2 percent. This is equivalent to an annual dividend of \$5.88 per share and marked the 62nd consecutive year of dividend increases. Other cash flows from financing activities may include various other items, such as cash paid associated with certain derivative instruments, distributions to or sales of noncontrolling interests, changes in cash overdraft balances, and principal payments for finance leases. Free Cash Flow (non-GAAP measure): Free cash flow and free cash flow conversion are not defined under U.S. generally accepted accounting principles (GAAP). Therefore, they should not be considered a substitute for income or cash flow data prepared in accordance with U.S. GAAP and may not be comparable to similarly titled measures used by other companies. The Company defines free cash flow as net cash provided by operating activities less purchases of property, plant and equipment. It should not be inferred that the entire free cash flow amount is available for discretionary expenditures. The Company defines free cash flow conversion as free cash flow divided by net income attributable to 3M. The Company believes free cash flow and free cash flow conversion are meaningful to investors as they are useful measures of performance and the Company uses these measures as an indication of the strength of the company and its ability to generate cash. The first quarter of each year is typically 3M's seasonal low for free cash flow and free cash flow conversion. Below find a recap of free cash flow and free cash flow conversion for 2019, 2018 and 2017. In 2019, free cash flow conversion was

Pipeline Parameters

Top-K Passages (k) 5

Embedding Model all-MiniLM-L6-v2 (Default Fast)

LLM Generation Backend Qwen2.5-1.5B-Instruct (Local HF)

Switching either model takes effect on your next query. The first query after switching loads that model into memory, so it runs slower once.

Chunking Strategy

Chunk Size: 512 tokens
Chunk Overlap: 64 tokens
Structure Preserved: Yes

Figure A.2 — Grounded, cited answer (cropped view).

A.3 — Honest abstention

Example query “What drove operating margin change as of FY2022 for 3M?” No retrieved passage directly answers the question, and the model responds “I cannot answer this from the provided documents” rather than fabricating a number. This is the behavior the citation-compliance prompt fix (§3.5) was designed to produce, shown here on a live, unseen query rather than an offline log line.

FinanceBench RAG

FINANCIAL RAG SYSTEM

Query Engine

Benchmark Analytics

PDF Store

Local Models Ready

Financial Query Console

What drove operating margin change as of FY2022 for 3M?

Ask RAG

SUGGESTED BENCHMARK QUESTIONS:

What is the FY2018 capital expenditure amount (in USD millions) for 3M?

What is the year end FY2018 net PPNE for 3M? Answer in USD billions.

Does 3M have a reasonably healthy liquidity profile based on its quick ratio for Q2 of FY2023?

What drove operating margin change as of FY2022 for 3M?

If we exclude the impact of M&A, which segment has dragged down 3M's overall growth in 2022?

Grounded Answer

Latency: 19.58s · all-MiniLM-L6-v2 · Qwen2.5-1.5B-Instruct · k=5

I cannot answer this from the provided documents.

Retrieved Context Passages used as Evidence:

11

3M_2016_10K (p. 1)

Match: 53.7%

are used in the manufacture, repair and maintenance of automotive, marine, aircraft and specialty vehicles. 3M is also a leading global supplier of precision grinding technology serving customers in the area of hard-to-grind precision applications in industrial, automotive, aircraft and cutting tools. 3M develops and produces advanced technical ceramics for demanding applications in the automotive, oil and gas, solar, industrial, electronics and defense industries. 3M's core filtration platform and will help generate new growth opportunities across the company. Operating income: · Operating income margins increased by 0.2 percentage points to 21.9 percent, helped by the combination of lower raw material costs and selective selling price increases. · Operating income margins were negatively impacted by 0.4 percentage points due to the \$42 million restructuring charge, plus an additional 0.3 percentage points penalty due to the Membrana acquisition. Safety and Graphics Business (18.8% of consolidated sales): Sales (millions) \$ 5,660 \$ 5,515 \$ 5,732 Sales change analysis: Organic local currency 2.2 % 2.4 % 5.4 % Acquisitions 4.0 2.6 - Divestitures (1.9) (0.4) - Translation (1.7) (3.4) (2.7) Total sales change 2.6 % (3.0)% 2.7 % Operating income (millions) \$ 1,390 \$ 1,395 \$ 1,394 Percent change 6.6 % 0.7 % 5.6 % Percent of sales 24.6 % 23.7 % 22.6 % The Safety and Graphics segment serves a broad range of markets that increase the safety, security and productivity of people, facilities and systems. Major product offerings include personal protection products, such as respiratory, hearing, eye and fall protection equipment; traffic safety and security products, including border and civil security solutions; commercial solutions, including commercial graphics sheeting and systems, architectural design solutions for surfaces, and cleaning and protection products for commercial establishments; and roofing granules for asphalt shingles. Year 2016 results: Sales in Safety and Graphics totaled \$5.7 billion, an increase of 2.6 percent in U.S. dollars. Organic local-currency sales increased 2.2 percent, and foreign currency translation reduced sales by 1.7 percent. Acquisitions added 4.0 percent, while divestitures reduced sales by 1.9 percent. On an organic local-currency sales basis: · Sales growth was led by roofing granules, which had a consistently strong year. · Commercial solutions and personal safety also showed positive growth. · Sales declined in traffic safety and security. · Sales increased 4 percent in Latin America/Canada, 3 percent in the United States and 2 percent in Asia Pacific, while sales in EMEA were flat. Acquisitions and divestitures (also discussed in Note 2): · Acquisition sales growth reflects the acquisition of Capital Safety in August 2015. Capital Safety is a leading global provider of fall protection equipment. · In the fourth quarter of 2015, 3M divested its license plate converting business in France and substantially all of its library systems business. In the first quarter of 2016, 3M divested the remainder of the library systems business. Operating income: · Operating income totaled \$1.4 billion, up 6.6 percent. · Operating income margins were 24.6 percent of sales, compared to 23.7 percent in 2015, benefiting from higher selling prices and lower raw material costs, plus productivity benefits related to fourth quarter 2015 restructuring actions. · Operating income margins were negatively impacted by margin dilution related to the Capital Safety acquisition, and divestiture impacts on margins.

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3M_2022_10K (p. 53)

Match: 52%

3M Company and Subsidiaries Notes to Consolidated Financial Statements NOTE 1. Significant Accounting Policies Consolidation: 3M is a diversified global manufacturer, technology innovator and marketer of a wide variety of products. All applicable subsidiaries are consolidated. All intercompany transactions are eliminated. As used herein, the term "3M" or "Company" refers to 3M Company and subsidiaries unless the context indicates otherwise. 3M deconsolidated the Anaro Entities in the third quarter of 2022. See additional information in Note 16. Basis of presentation: Certain amounts in the prior years' consolidated financial statements have been reclassified to conform to the current year presentation. Effective in the first quarter of 2022, 3M made changes in the measure of segment operating performance used by 3M's chief operating decision maker—impacting 3M's disclosed measure of segment profit/loss (business segment operating income). See additional information in Note 19. 3M's disclosed disaggregated revenue was also updated as a result of the changes in segment reporting. See additional information in Note 2. Information provided herein reflects the impact of these changes for all periods presented. Foreign currency translation: Local currencies generally are considered the functional currencies outside the United States. Exceptions include 3M's subsidiaries in Argentina and, beginning in the second quarter of 2022, in Turkey, the economy of which also became highly inflationary. The operating income and balances of underlying net monetary assets denominated in Turkish lira are not material to 3M. The financial statements of these subsidiaries are remeasured as if their functional currency is that of their parent. Assets and liabilities for operations in local-currency environments are translated at month-end exchange rates of the period reported. Income and expense items are translated at average monthly currency exchange rates in effect during the period. Cumulative translation adjustments are recorded as a component of accumulated other comprehensive income (loss) in shareholders' equity. Use of estimates: The preparation of financial statements in conformity with U.S. generally accepted

Figure A.3 — Honest abstention on an unanswerable query (cropped view).

Finance Bench RAG — NLP Graded Project A | Page 14

A.4 — Benchmark analytics tab

Live summary cards (best score per metric across all 20 configurations), an interactive comparison chart, and the full experimental matrix, all read directly from results/comparison.csv — the same numbers reported in Section 5.

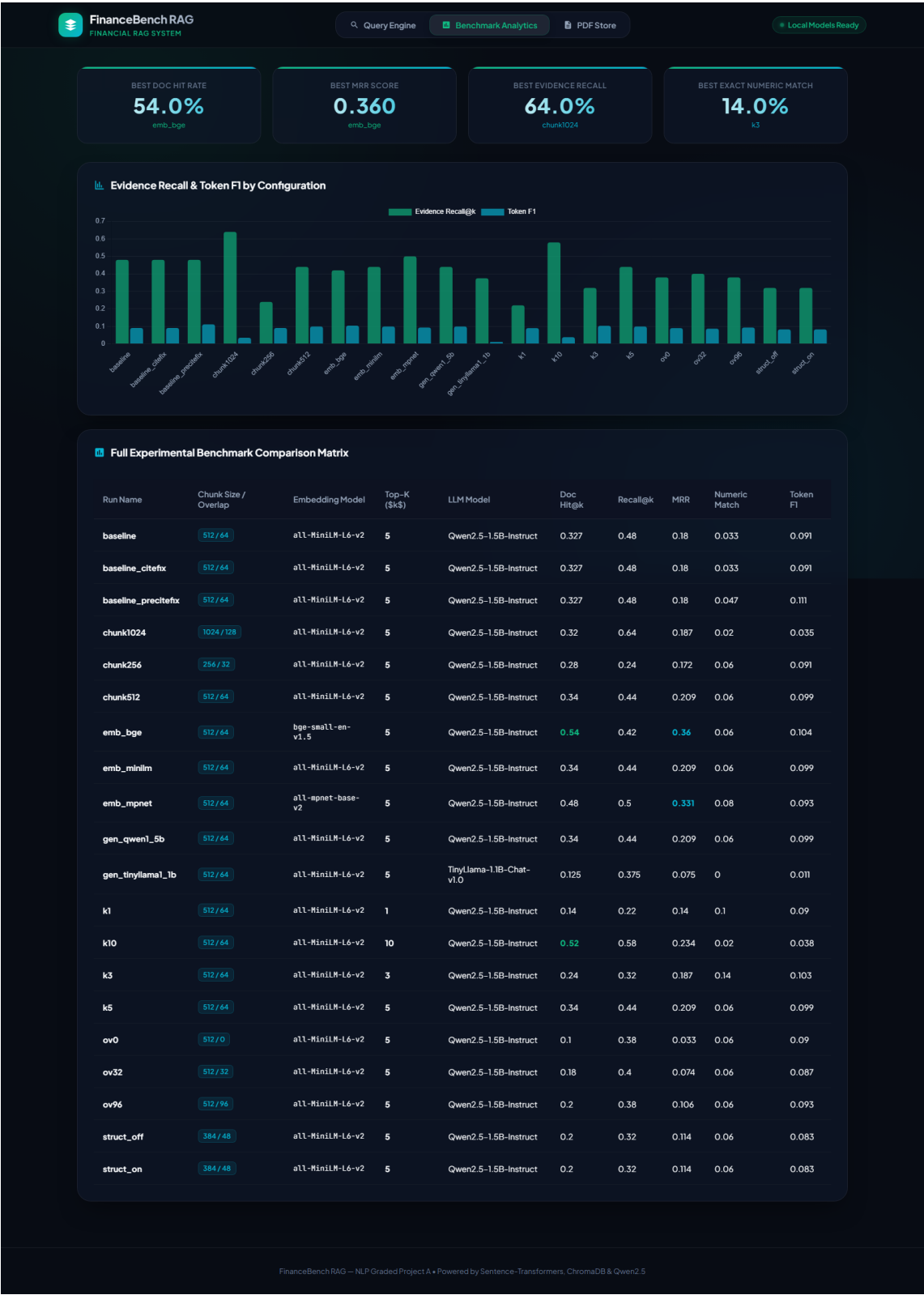


Figure A.4 — Benchmark analytics dashboard.

A.5 — PDF store / corpus browser

A live listing confirming the 263-document corpus described in Section 2, with per-document file type and size.

The screenshot displays the 'FinanceBench RAG' interface, specifically the 'PDF Store' tab. The header includes a logo, the text 'FINANCIAL RAG SYSTEM', and navigation links for 'Query Engine', 'Benchmark Analytics', and 'PDF Store'. A status indicator 'Local Models Ready' is visible in the top right. The main content area is titled 'Indexed SEC Filing Knowledge Base (263 PDFs)' and shows 'Showing the first 50 of 263 — full list available via /api/documents.' Below this, a grid of 20 document cards is displayed, each representing a PDF file with its name and size. The documents are organized by company and year, including 3M, AES, AMAZON, AMCOR, AMD, AMERICANEXPRESS, AMERICANWATERWORKS, and APPLE. The footer of the interface reads 'FinanceBench RAG — NLP Graded Project A • Powered by Sentence-Transformers, ChromaDB & Qwen2.5'.

Document Name	Size
3M_2015_10K.pdf	10-K • 1.11 MB
3M_2016_10K.pdf	10-K • 1.38 MB
3M_2017_10K.pdf	10-K • 1.31 MB
3M_2018_10K.pdf	10-K • 1.19 MB
3M_2019_10K.pdf	10-K • 1.17 MB
3M_2020_10K.pdf	10-K • 1.04 MB
3M_2021_10K.pdf	10-K • 1.09 MB
3M_2022_10K.pdf	10-K • 1.58 MB
3M_2023Q2_10Q.pdf	10-Q • 5.04 MB
AES_2015_10K.pdf	10-K • 1.94 MB
AES_2016_10K.pdf	10-K • 2.8 MB
AES_2017_10K.pdf	10-K • 6.36 MB
AES_2018_10K.pdf	10-K • 4.05 MB
AES_2019_10K.pdf	10-K • 7.53 MB
AES_2020_10K.pdf	10-K • 3.46 MB
AES_2021_10K.pdf	10-K • 2.4 MB
AES_2022_10K.pdf	10-K • 4.27 MB
AMAZON_2015_10K.pdf	10-K • 0.57 MB
AMAZON_2016_10K.pdf	10-K • 0.57 MB
AMAZON_2017_10K.pdf	10-K • 0.8 MB
AMAZON_2018_10K.pdf	10-K • 0.61 MB
AMAZON_2019_10K.pdf	10-K • 0.59 MB
AMAZON_2020_10K.pdf	10-K • 0.55 MB
AMAZON_2021_10K.pdf	10-K • 0.95 MB
AMAZON_2022_10K.pdf	10-K • 0.68 MB
AMCOR_2019_10K.pdf	10-K • 1.33 MB
AMCOR_2020_10K.pdf	10-K • 7.98 MB
AMCOR_2021_10K.pdf	10-K • 1.07 MB
AMCOR_2022_10K.pdf	10-K • 1.01 MB
AMCOR_2022_8K_dated-2022...	8-K • 1.67 MB
AMCOR_2022_8K_dated-2022...	8-K • 0.14 MB
AMCOR_2023_10K.pdf	10-K • 1.06 MB
AMCOR_2023Q2_10Q.pdf	10-Q • 0.43 MB
AMCOR_2023Q4_EARNINGS.pdf	8-K • 0.23 MB
AMD_2022_10K.pdf	10-K • 4.89 MB
AMERICANEXPRESS_2022_10K...	10-K • 2.32 MB
AMERICANWATERWORKS_2015...	10-K • 1.37 MB
AMERICANWATERWORKS_2016...	10-K • 1.02 MB
AMERICANWATERWORKS_2017...	10-K • 1.2 MB
AMERICANWATERWORKS_2018...	10-K • 1.46 MB
AMERICANWATERWORKS_2019...	10-K • 1.73 MB
AMERICANWATERWORKS_2020...	10-K • 1.37 MB
AMERICANWATERWORKS_2021...	10-K • 1.65 MB
AMERICANWATERWORKS_2022...	10-K • 2.0 MB
AMERICANWATERWORKS_2023...	10-Q • 0.62 MB
APPLE_2015_10K.pdf	10-K • 0.67 MB
APPLE_2016_10K.pdf	10-K • 0.82 MB
APPLE_2017_10K.pdf	10-K • 0.9 MB
APPLE_2018_10K.pdf	10-K • 0.78 MB
APPLE_2019_10K.pdf	10-K • 0.76 MB

Figure A.5 — Indexed SEC-filing knowledge base.